Coprime Ramsey Numbers: Exact Values and Structural Properties

Alexander Towell*

Abstract

We introduce the *coprime Ramsey number* $R_{\text{cop}}(k)$, defined as the minimum n such that every 2-coloring of the edges of the coprime graph on $\{1, \ldots, n\}$ contains a monochromatic complete subgraph K_k. Using a combination of incremental extension and SAT solving, we establish $R_{\text{cop}}(2) = 2$, $R_{\text{cop}}(3) = 11$, and present strong computational evidence that $R_{\text{cop}}(4) = 59$. We compute the 3-color triangle number $R_{\text{cop}}(3; 3) = 53$ and determine exact values for path, cycle, asymmetric, and Gallai variants. All known values of $R_{\text{cop}}(k)$ and $R_{\text{cop}}(3; 3)$ are prime—we state this as a conjecture.

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1 Introduction

The *coprime graph* $G_{\text{cop}}(n)$ has vertex set $[n] = \{1, 2, \ldots, n\}$ with $\{i, j\}$ an edge whenever $\gcd(i, j) = 1$. This graph, studied by Erdős and Sárközy [2] and others [5, 1], encodes number-theoretic structure in a combinatorial object: its edge density converges to $6/\pi^2 \approx 0.608$, its clique number equals $1 + \pi(n)$ (where π is the prime-counting function), and its independence number is $\lfloor n/2 \rfloor$.

We propose a Ramsey-theoretic study of this graph. Classical Ramsey numbers $R(k, k)$ concern the complete graph K_n; replacing K_n with $G_{\text{cop}}(n)$ yields a problem where Ramsey-type forcing must contend with the arithmetic structure of coprimality. This appears to be a new line of investigation.

Definition 1.1. The *coprime Ramsey number* $R_{\text{cop}}(k)$ is the minimum n such that every 2-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic K_k. More generally, $R_{\text{cop}}(k; c)$ is the minimum n such that every c-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic K_k.

Since $E(G_{\text{cop}}(n)) \subseteq E(K_n)$, any avoiding coloring of $G_{\text{cop}}(n)$ extends (by coloring non-coprime edges arbitrarily) to an avoiding coloring of K_n. Thus $R_{\text{cop}}(k) \geq R(k, k)$, and forcing monochromatic cliques in the sparser coprime graph requires larger n.

2 Exact values

Theorem 2.1. $R_{\text{cop}}(2) = 2$.

Proof. Since $\gcd(1, 2) = 1$, the graph $G_{\text{cop}}(2)$ has a single edge. Any 2-coloring assigns it one color, producing a monochromatic K_2. □

*Department of Computer Science, Southern Illinois University Edwardsville. Email: `atowell@siue.edu`. ORCID: 0000-0001-6443-9897.

Theorem 2.2. $R_{\text{cop}}(3) = 11$.

Proof. By exhaustive incremental extension. Among all 2^{21} edge 2-colorings of $G_{\text{cop}}(8)$ (21 coprime edges), exactly 36 avoid monochromatic triangles. We extend these through vertices 9, 10, 11: for each new vertex v, the coprime edges from v to $\{1, \ldots, v-1\}$ are enumerated, and every 2-coloring of these new edges is tested against each surviving avoiding coloring.

n	Coprime edges	New edges at n	Avoiding colorings
8	21	—	36
9	27	6	36
10	31	4	156
11	41	10	0

The increase from 36 to 156 at $n = 10$ occurs because $10 = 2 \times 5$ is coprime to only 4 predecessors, imposing weak constraints. The collapse to 0 at $n = 11$ is driven by 11 being prime: it is coprime to all of $\{1, \ldots, 10\}$, creating 10 new edges and far too many triangles for any 2-coloring to avoid. □

Remark 2.3. $R_{\text{cop}}(3)/R(3,3) = 11/6 \approx 1.83$. If $G_{\text{cop}}(n)$ behaved as a random graph of density $6/\pi^2$, one would predict $R_{\text{cop}}(3) \approx 10$. The slight excess reflects multiplicative correlations among missing edges: multiples of a prime p form correlated non-adjacencies in $G_{\text{cop}}(n)$ [2].

Theorem 2.4. $R_{\text{cop}}(4) = 59$, *subject to the caveat below.*

Evidence. The lower bound $R_{\text{cop}}(4) \geq 59$ is established by SAT solving: for every $n \leq 58$, the Glucose4 solver [3] finds a 2-coloring of $E(G_{\text{cop}}(n))$ with no monochromatic K_4 (returning SAT). For the upper bound, we use an extension-based argument: 100 distinct avoiding colorings of $E(G_{\text{cop}}(58))$ are extracted via SAT with blocking clauses, and for each one, the SAT instance encoding the extension to $n = 59$ (coloring the 58 new edges from the prime vertex 59 to all of $\{1, \ldots, 58\}$) is UNSAT. Since 59 is prime, it is coprime to every predecessor, and the resulting constraint density (clause-to-variable ratio ≈ 109) far exceeds the random 3-SAT UNSAT threshold (≈ 4.27).

Caveat. The extension check covers 100 seeds, not all avoiding colorings at $n = 58$. A rigorous proof would require either (a) a direct UNSAT certificate for the full $n = 59$ SAT instance (1,085 variables, ∼118,000 clauses—which timed out after 30+ minutes with CDCL solvers), or (b) enumeration of *all* avoiding colorings at $n = 58$ and verification that none extend. We regard the evidence as very strong but acknowledge this gap.

k	$R_{\text{cop}}(k)$	$R(k,k)$
2	2	2
3	11	6
4	59*	18

*Strong computational evidence; see caveat above.

Theorem 2.5. $R_{\text{cop}}(3;3) = 53$.

Proof. By SAT solving: for $n \leq 52$, a 3-coloring of $E(G_{\text{cop}}(n))$ avoiding monochromatic triangles exists (SAT). At $n = 53$, the instance is UNSAT. The 3-color encoding uses $c - 1 = 2$ Boolean variables per edge with at-most-one constraints and, for each triangle and each color, forbids monochromatic assignment. □

3 Variant table

We define and compute several Ramsey-type variants on $G_{\text{cop}}(n)$. All values are exact, determined by complete SAT search at each n.

Definition 3.1. Let $c \geq 2$ be the number of colors and $k \geq 2$ the target structure size.

1. $P_{\text{cop}}(k)$: minimum n such that every 2-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic path on k edges.

2. $C_{\text{cop}}(k)$: minimum n such that every 2-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic cycle on k edges.

3. $R_{\text{cop}}(s,t)$: minimum n such that every 2-coloring of $E(G_{\text{cop}}(n))$ has a monochromatic K_s in color 0 or K_t in color 1 (asymmetric coprime Ramsey).

4. $\text{GR}_{\text{cop}}(k;c)$: minimum n such that every c-coloring of $E(G_{\text{cop}}(n))$ contains a monochromatic or rainbow K_k (Gallai coprime Ramsey; requires $c \geq 3$).

Table 1: Computed coprime Ramsey variants.

Variant	**Values**	**Classical counterpart**
$P_{\text{cop}}(k)$, $k = 3, \ldots, 8$	5, 7, 9, 10, 13, 13	—
$C_{\text{cop}}(k)$, $k = 3, \ldots, 6$	11, 8, 13, 11	—
$R_{\text{cop}}(2,3)$	3	$R(2,3) = 6$
$R_{\text{cop}}(2,4)$	5	$R(2,4) = 9$
$R_{\text{cop}}(3,4)$	19	$R(3,4) = 9$
$R_{\text{cop}}(3;3)$	53	$R(3,3,3) = 17$
$\text{GR}_{\text{cop}}(3;3)$	29	$\text{GR}(3;3) = 17$

Remark 3.2. Several patterns emerge from Table 1.

1. *Path versus clique.* $P_{\text{cop}}(k) \ll R_{\text{cop}}(k)$: paths are much easier to force. The values plateau at 13 for $k = 7, 8$ because $n = 13$ is prime, making $G_{\text{cop}}(13)$ dense enough to force long monochromatic paths.

2. *Even–odd cycle gap.* $C_{\text{cop}}(4) = 8 < C_{\text{cop}}(3) = 11$ and $C_{\text{cop}}(6) = 11 < C_{\text{cop}}(5) = 13$: even-length cycles are easier to force.

3. *Multi-color growth.* $R_{\text{cop}}(3;3)/R_{\text{cop}}(3;2) = 53/11 \approx 4.8$, compared with the classical $R(3,3,3)/R(3,3) = 17/6 \approx 2.8$. The coprime graph's structure amplifies multi-color avoidance.

4. *Coprime amplification.* For every symmetric variant, the coprime version exceeds its classical counterpart: $R_{\text{cop}}(3) > R(3,3)$, $R_{\text{cop}}(3;3) > R(3,3,3)$, $\text{GR}_{\text{cop}}(3;3) > \text{GR}(3;3)$.

4 Structural properties

Proposition 4.1. $\omega(G_{\text{cop}}(n)) = 1 + \pi(n)$.

Proof. The set $\{1\} \cup \{p \leq n : p \text{ prime}\}$ is a clique of size $1 + \pi(n)$. For the upper bound: in any pairwise coprime set $S \subseteq \{2, \ldots, n\}$, each element's prime factorization uses a disjoint set of primes, so $|S| \leq \pi(n)$. □

Proposition 4.2. *$G_{\text{cop}}(n)$ is weakly perfect: $\chi(G_{\text{cop}}(n)) = \omega(G_{\text{cop}}(n)) = 1 + \pi(n)$.*

This follows from the observation that the partition $\{1\}, \{2, 4, 6, \ldots\}, \{3, 9, 27, \ldots\}, \{5, 25, \ldots\}, \ldots$ into classes indexed by the smallest prime factor gives a proper $(1 + \pi(n))$-coloring. Perfectness of related coprime-type graphs on group elements was established by Syarifudin and Wardhana [7].

Proposition 4.3. $\alpha(G_{\text{cop}}(n)) = \lfloor n/2 \rfloor$.

Proof. The even numbers $\{2, 4, 6, \ldots, 2\lfloor n/2 \rfloor\}$ form an independent set of size $\lfloor n/2 \rfloor$ (every pair shares factor 2). No larger independent set exists: in any set S of $\lfloor n/2 \rfloor + 1$ elements of $[n]$, by pigeonhole two share the same value of $\lceil v/2 \rceil$, forcing a coprime pair $(2k - 1, 2k)$. □

Conjecture 4.4 (Primality conjecture)**.** *All diagonal coprime Ramsey numbers $R_{\text{cop}}(k)$ are prime. More generally, $R_{\text{cop}}(k; c)$ is prime for all k, c.*

Evidence. $R_{\text{cop}}(2) = 2$, $R_{\text{cop}}(3) = 11$, $R_{\text{cop}}(4) = 59$, and $R_{\text{cop}}(3; 3) = 53$ are all prime. The heuristic explanation is that prime values of n maximize the coprime density of $G_{\text{cop}}(n)$—a prime p is coprime to all of $\{1, \ldots, p-1\}$—so the transition from SAT to UNSAT preferentially occurs at primes, where the addition of a universal coprime vertex creates the maximal number of new constraints.

5 Methodology

All computations use the PySAT toolkit [4] with the Glucose4 [3] CDCL solver. The methodology varies by problem size.

Small instances ($R_{\text{cop}}(3)$, path, cycle, Gallai variants). For each n, we encode the avoidance problem as a SAT instance: one Boolean variable per coprime edge ($x_e = \top$ for color 0, $x_e = \bot$ for color 1), two clauses per target substructure (forbidding all-color-0 and all-color-1). The solver returns SAT or UNSAT. The value $R_{\text{cop}}(3) = 11$ is also verified by the exact incremental extension method of Theorem 2.2, which enumerates *all* avoiding colorings at each n and is thus a complete proof.

Large instances ($R_{\text{cop}}(4)$). The incremental SAT approach determines $R_{\text{cop}}(4) \geq 59$: Glucose4 returns SAT for all $n \leq 58$, with solving times under 1 second per instance. For the upper bound, we extract diverse avoiding colorings at $n = 58$ via SAT with blocking clauses, then check extension to $n = 59$ by fixing the base coloring as unit-propagation assumptions. All 100 tested seeds produce UNSAT extensions. As noted in Theorem 2.4, this is not a complete UNSAT proof for $n = 59$.

Multi-color variants ($R_{\text{cop}}(3; 3)$, $\text{GR}_{\text{cop}}(3; 3)$). For c colors, each edge has $c-1$ Boolean variables with at-most-one pairwise exclusion constraints. For each clique and each color, a clause forbids monochromatic assignment. The Gallai variant additionally forbids rainbow cliques via all-distinct permutation clauses. These instances are within Glucose4's reach for the parameter ranges in Table 1.

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Code and data: `github.com/queelius/erdos-research`

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